

Rect

`x = 300
height = 400
isTop: true
width: 20
moveLeft() {
 x = x - 1
}`

`x = 500
height = 200
isTop: true
width: 10
moveLeft() {
 x = x - 1
}`

Classes

1. x
2. height
3. isTop
4. width

`moveLeft() {
 x--
}`

bird

`x = 400`

Rect

`x = 300
height = 400
isTop: true
width: 20
moveLeft() {
 x = x - 1
}`

`x = 500
height = 200
isTop: true
width: 10
moveLeft() {
 x = x - 1
}`

class Rectangle {

1. x
2. height
3. isTop
4. width

`moveLeft() {
 x--
}`

bird

`x = 400`

Rect

Rectangle 1:
 $x = 300$
 $height = 400$
 $isTop: true$
 $width: 20$
 $moveLeft() \{$
 $x = x - 1$
 $\}$

Rectangle 2:
 $x = 500$
 $height = 200$
 $isTop: true$
 $width: 10$
 $moveLeft() \{$
 $x = x - 1$
 $\}$

Rectangle 3:
 $x = 400$
 $height = 250$
 $isTop: true$
 $width: 10$
 $moveLeft() \{$
 $x = x - 1$
 $\}$

Code Snippet:
 $let r1 = new Rectangle(10, 10, "red")$
 $console.log(r1.area())$

Class Definition:
 $class Rectangle \{$
 $constructor(width, height, color) \{$
 $this.width = width;$
 $this.height = height;$
 $this.color = color;$
 $\}$
 $area() \{$
 $const area = this.width * this.height;$
 $return area;$
 $\}$
 $paint() \{$
 $console.log('Painting with color ' + this.color);$
 $\}$
 $\}$

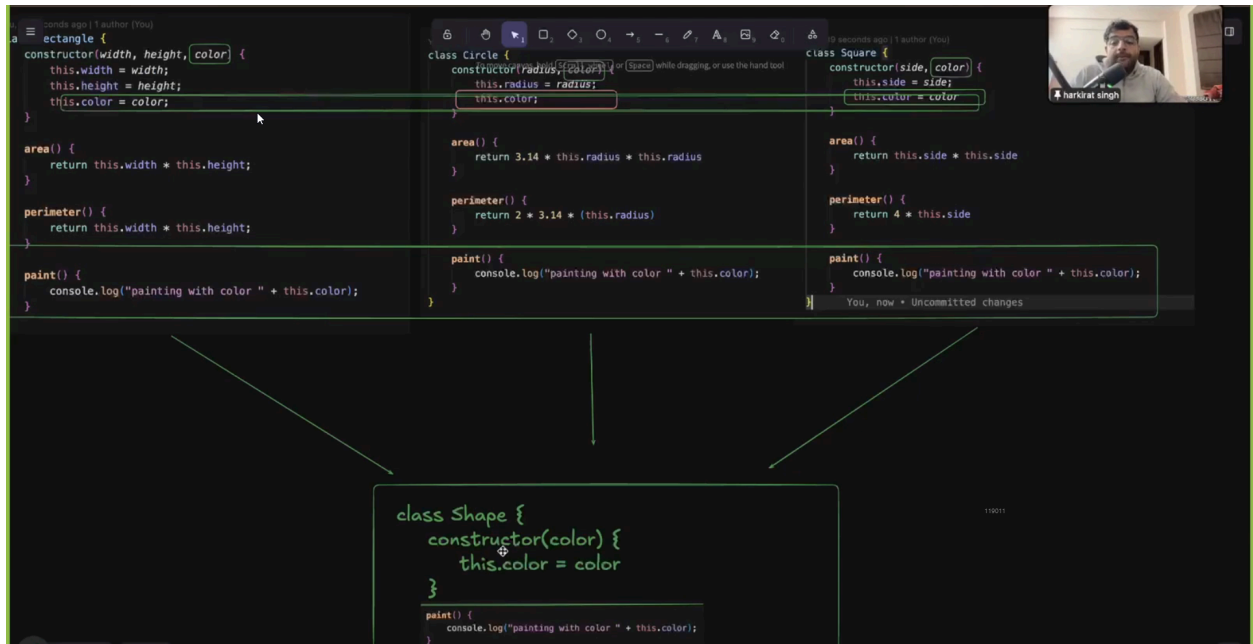
Inheritance

Rectangle Class:
 $class Rectangle \{$
 $constructor(width, height, color) \{$
 $this.width = width;$
 $this.height = height;$
 $this.color = color;$
 $\}$
 $area() \{$
 $return this.width * this.height;$
 $\}$
 $perimeter() \{$
 $return this.width * this.height;$
 $\}$
 $paint() \{$
 $console.log("painting with color " + this.color);$
 $\}$
 $\}$

Circle Class:
 $class Circle \{$
 $constructor(radius, color) \{$
 $this.radius = radius;$
 $this.color = color;$
 $\}$
 $area() \{$
 $return 3.14 * this.radius * this.radius;$
 $\}$
 $perimeter() \{$
 $return 2 * 3.14 * (this.radius);$
 $\}$
 $paint() \{$
 $console.log("painting with color " + this.color);$
 $\}$
 $\}$

Square Class:
 $class Square \{$
 $constructor(side, color) \{$
 $this.side = side;$
 $this.color = color;$
 $\}$
 $area() \{$
 $return this.side * this.side;$
 $\}$
 $perimeter() \{$
 $return 4 * this.side;$
 $\}$
 $paint() \{$
 $console.log("painting with color " + this.color);$
 $\}$
 $\}$

DRY rule - dont repeat yourself



Promise

```
function callback() {
  console.log("hi there");
}
```

```
setTimeout(callback, 5 * 1000);
```

```
let ctr = 0;
for (let i = 0; i < 1000; i++) {
  ctr = ctr + i;
}
```

```
console.log(ctr);
```